

Navigating the
**QUANTUM
MAZE**

*A High School Guide to the
Strangest Computer Science*

Every slide is a new corridor. Let's explore →

THE FIRST CORRIDOR: CLASSICAL COMPUTERS

Before entering the quantum maze, we need to understand the straight hallways we already know.

THE CLASSICAL BIT

A classical computer thinks in bits — tiny switches that are either ON (1) or OFF (0). Every app, game, and website you use runs on billions of these simple yes-or-no decisions happening incredibly fast.

THE MAZE ANALOGY

Imagine a maze where you can only walk one path at a time. You reach a fork: go left or right — pick one. A classical computer solves mazes this way, trying each route one by one until it finds the exit.

WHAT MAKES QUANTUM DIFFERENT?

The maze just got a whole lot stranger.

QUBITS

The Quantum Walker

Unlike a classical bit (0 or 1), a qubit can exist in a blend of both states simultaneously — like a maze-walker who hasn't decided which fork to take yet.

QUANTUM GATES

The Maze Architect

Quantum gates are operations that reshape the maze itself — rotating, flipping, and combining paths in ways that classical logic gates cannot. They redesign reality.

MEASUREMENT

The Collapse

When you observe a qubit, it 'chooses' a path. Like shining a flashlight in the maze — the walker is found in one corridor, and all other ghostly paths vanish instantly.

SUPERPOSITION

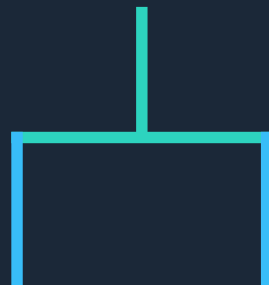
The Walker Who Takes Every Path at Once

In a classical maze, you pick one corridor and hope for the best. But a quantum walker in superposition doesn't choose — it explores EVERY possible path simultaneously.

Picture yourself as a ghost who splits into copies at every fork, flowing through the entire maze at once. Each copy carries a probability of being the 'real' you.

When someone finally looks (measures), all copies collapse into a single location. This is the secret power of quantum computing: instead of checking one answer at a time, a quantum computer checks many in parallel.

MAZE FORK



Classical:
Pick ONE



Quantum :
Take BOTH



TAKE THE QUIZ

INTERFERENCE: WHERE PATHS COLLIDE

The maze helps you by erasing wrong answers

Think of ripples in water. When two waves meet, they can amplify each other (constructive interference) or cancel each other out (destructive interference).

In the quantum maze, your ghost-copies carry wave-like amplitudes. Correct paths reinforce each other, growing stronger. Wrong paths interfere destructively, fading to nothing.

Quantum algorithms are carefully designed so the right answer's paths add up while wrong answers cancel themselves out — like a maze that actively erases its own dead ends.

CONSTRUCTIVE



DESTRUCTIVE



The quantum maze deletes its own dead ends.

ENTANGLEMENT

Two corridors, one destiny

Two qubits become entangled when their fates are linked — measuring one INSTANTLY reveals the state of the other, no matter how far apart they are. Einstein called it "spooky action at a distance." In maze terms: two walkers in separate corridors are mysteriously synchronized.

QUBIT A

The walker enters the LEFT corridor.
Before measurement: could be at any junction.
State is uncertain — 0, 1, or both at once.



QUBIT B

The walker enters the RIGHT corridor.
The moment Qubit A is found, Qubit B is INSTANTLY found at its linked position.

QUANTUM ALGORITHMS

Smart strategies for navigating the maze at quantum speed

GROVER'S ALGORITHM

Finding a needle in a haystack

Imagine the maze has one million rooms and the exit is hidden in just one. A classical computer checks rooms one by one — up to a million tries.

Grover's algorithm sends quantum walkers through ALL rooms at once, using interference to amplify the correct room's signal. It finds the exit in roughly 1,000 steps instead of 1,000,000.

SHOR'S ALGORITHM

Cracking codes by reshaping the maze

Modern encryption relies on the fact that factoring huge numbers is astronomically slow for classical computers.

Shor's algorithm transforms this problem into a maze with hidden periodic patterns, then uses quantum interference to reveal those patterns exponentially faster. This could one day break most internet encryption.

REAL-WORLD QUANTUM COMPUTERS

The maze is being built right now — here's who's constructing it

IBM

Eagle processor with 127+ qubits. Uses superconducting circuits cooled near absolute zero — like building a maze inside a freezer colder than outer space.

GOOGLE

Achieved 'quantum supremacy' in 2019 — Sycamore solved a problem in 200 seconds that would take a supercomputer 10,000 years. The maze, outrunning all classical explorers.

MICROSOFT

Betting on topological qubits — a radically different maze architecture inherently resistant to errors, like building walls that repair themselves.

STARTUPS

IonQ uses trapped ions, Rigetti builds cloud chips, PsiQuantum pursues photonic qubits — each designing their own unique style of quantum maze.



TAKE THE QUIZ

CHALLENGES: THE MAZE FIGHTS BACK

Building a quantum computer is like constructing a maze out of fog

DECOHERENCE

The walls dissolve

Qubits are incredibly fragile. Heat, vibration, or a stray photon causes the quantum maze to collapse. That's why quantum chips operate at -459°F — colder than outer space.

ERROR CORRECTION

Rebuilding walls as they crumble

Thousands of physical qubits create one reliable 'logical qubit.' Imagine needing 1,000 workers to maintain a single maze wall. This is the biggest engineering challenge.

SCALABILITY

Making the maze bigger

Today: hundreds of qubits. Useful applications need millions. Each new corridor makes every existing one harder to maintain — like extending a maze on a vibrating table.

YOU'VE NAVIGATED THE QUANTUM MAZE

From classical bits to qubits, from single paths to superposition, from isolated corridors to entangled passages — you've traveled through the fundamental ideas that make quantum computing one of the most exciting frontiers in science and technology.

*The maze of quantum computing is still being built.
Maybe one day, you'll help design its next corridor.*

Thank you for exploring with us.